RESEARCH LOGBOOK · DELIVERABLE B

WHITEBOX

Gauge-Invariant Certification and Homological Reconstruction of Learned Representations in Overparameterized Neural Networks for Provable Mechanistic Interpretability

$$g = J^T J \cdot H^1(M; \mathbb{Z}) \to S^1 \cdot Aut(D_g) \supseteq \mathbb{Z}_{113} \cdot M \mapsto M / G$$



PROJECT TITLE	WHITEBOX — opening the black box of neural-network learning
STUDENTS	Arjun Paradkar · Reva Paradkar
SCHOOL	Independent — no institutional affiliation
CATEGORY	Natural & Applied Science (NAS)
COMPETITION	International AI Innovation Olympiad — AI for Science, Global Final Round
PERIOD COVERED	4 May 2026 → 22 September 2026

A chronological record of the project from its first draft to the current experimental run: what was done, with what tools, what was found, and what changed as a result — including every result that failed. Entries are never revised; corrections appear as later entries. Dates and clock times come from file timestamps, git history and run logs wherever those exist; dates marked ~ have no file record and are recalled to within a few days. Compiled 22 September 2026.

The project in one page

"Black box" names seven different failures: we do not know **what** a network learned (P1), **why** it decided (P2), **when bias entered** (P3), **what it does not know** (P4), **where it will fail** (P5), whether it **memorised or understood** (P6), or **how its representations formed** (P7). WHITEBOX spent five months trying to put all seven on one mathematical object — and changing the object each time the evidence demanded it.

20	**4**	**4,470+**	**7**	**135**	**213**
weeks, first draft to Run 4	mathematical formulations, each retired or absorbed by the next	networks trained and analysed (3,648 in one battery)	pre-registered runs executed in 3 days	commits in the archived repos; 12,300 lines in the live one	unit tests on the final library, instruments verified on planted truth

I · MAY – JULY

$SV = W_2(D_t, D_{t-1})$

The **topological film**: persistent homology of activation clouds across training. Structural velocity, topological complexity TC = $\Sigma(d_i - b_i)^2$, feature lifetime.

II · AUGUST

$TC(\lambda X) = \lambda^2\, TC(X)$

The **controls**. A shuffle null shows the headline 2,900× "collapse" was activation scale, not topology. Pre-registration becomes the method.

III · SEPTEMBER

$G = \prod S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}$

The **gauge** programme: certify explanations against the network's own symmetry group. Trivial-stabiliser proposition; K/N = 1.

IV · 19 – 21 SEPT

$g = J^T J, \quad M \mapsto M/G$

The **pullback metric** and **understanding as quotient**: a network that generalises carries a discoverable group; one that memorises does not.

Timeline at a glance

WHEN	STAGE	DATE ESTABLISHED FROM
4 May	Formulation: seven problems, the topological film, five hypotheses; first paper draft	draft PDF timestamp
6 – 14 May	First implementation and results (ResNet-18, TinyGPT); conference-style drafts	source + log + PDF timestamps
28 June	Version 1 at scale: GPT-2 124M bias intervention, six-problem battery, competition paper, Quad Chart	run directories 055217 / 151432; PDFs
2 – 29 July	Figures regenerated from raw arrays; NAS paper v2; US national round; literature check; rebuilt paper	PDF timestamps (lit. check ~)
12 – 27 Aug	Code, results and paper published on GitHub; explainer deck	repo + file timestamps
28 – 30 Aug	Library rebuild with nulls — most v1 claims retired. Literature campaign. E1–E6 programme; "The Structure of Understanding"	git history (15 commits)
1 – 17 Sept	Gauge-certification sprint: handbook, 3,648-network battery, K/N = 1 tautology, GPT-2 circuits on gauge orbits, mathematical write-ups	git history (120 commits), pause docs
19 Sept	Reset to a Riemannian theory of representations; 10-hour specification	archive note; specification PDF
19 – 22 Sept	Runs 1 – 7: pullback geometry → Morse theory → the symmetry group → the tower of quotients → registration → orbit-closure → the extractor. Final paper and backboard	run logs with clock times

How this logbook was assembled. An earlier compilation gave approximate dates from memory. For this version every entry was checked against the project archive — file timestamps, two git histories, run directories named by time, and the hour-by-hour run logs of 19–21 September. Where record and recollection disagreed the record is used: the project began in early May, not late June, and 28 June was one long day rather than a week. Durations are clock times where a log exists and estimates ("est.") otherwise. An AI coding agent (Claude Code, Anthropic) served as the execution agent and is named under *Methods & tools* wherever it applies; direction, hypotheses, specifications and the decisions under *Adjustments* are the authors'.

PART I · MAY – JULY 2026

The topological film

Persistent homology of activations across training as a single instrument for all seven problems.

4 MAY 2026

Formulation: seven problems, one film

Duration ~4 h (est.)
Record: draft “v3”, 4 May 18:59

RESEARCH ACTIVITIES

Defined the target: the black box as seven distinct problems, P1–P7. Proposed that the persistent homology of activation point clouds, tracked across training epochs — the **topological film** — reads structure that the loss curve cannot. Wrote five hypotheses: H1 a structural phase transition invisible to loss; H2 concept crystallisation ordered by difficulty; H3 bias appears in topology before it appears in outputs; H4 a memorisation signature; H5 cross-modal universality.

METHODS & TOOLS

Three metrics on persistence diagrams $D_t = \{(b_i, d_i)\}$: Structural Velocity $\mathrm{SV}(t) = W_2(D_t, D_{t-1})$; Topological Complexity $\mathrm{TC} = \Sigma_i (d_i - b_i)^2$; Feature Persistence Lifetime. Vietoris–Rips filtration (Ripser), Wasserstein distances (persim), PCA to 30 dimensions fitted at epoch 0.

PROCESS NOTES

The seven-problem framing was chosen because no existing method treats the seven failures as one object. TC was chosen because squaring lifetimes weights long-lived features — a choice that would matter in August.

STAGE OUTCOMES

A written framework and a draft paper (“Opening the Black Box of Neural Network Learning Through Persistent Homology of Activation Dynamics”).

ADJUSTMENTS

Next: implement, and test every hypothesis on a vision and a language model.

6 – 7 MAY 2026 · REVISION 14 MAY

First implementation and first results

Session 6 May ~22:20 → 7 May 00:20
Record: source, logs, five draft PDFs

RESEARCH ACTIVITIES

Wrote the core (`whitebox.py`) and three experiments — vision, language, memorisation — with a watchdog to chain them. Generated the first results report and four figures; produced conference-style drafts the same night and a revised draft on 14 May.

METHODS & TOOLS

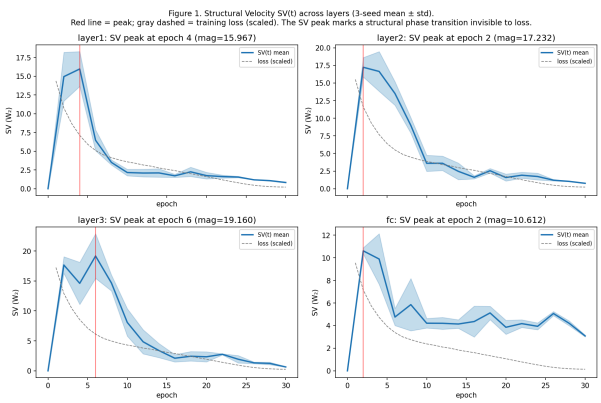
PyTorch, Ripser, persim. ResNet-18 on CIFAR-10 (3 seeds, clean vs 100% label noise); a from-scratch 6-layer TinyGPT (11.4 M parameters) standing in for GPT-2; a 200-sentence WinoBias-style probe. One laptop GPU (RTX 5070, 8 GB), AMP fp16.

```
# whitebox.py, 6 May 2026 — the two metrics everything started from
def structural_velocity(D_t, D_tm1) → float:
    ...
    return float(wasserstein(A, B))                    # SV = W2(D_t, D_{t-1})

def topological_complexity(D) → float:
    return float(np.sum((D[:, 1] - D[:, 0]) ** 2))     # TC = Σ (death - birth)²
```

STAGE OUTCOMES

H1 supported: every layer shows an SV peak with no matching feature in the loss; cross-seed $\rho = 0.931$. H2 trend-level: $\rho = 0.585$, $p = 0.076$. H3 mixed: a 2-epoch lead at layer 0 only. H4: TC 1.46 (clean) vs 0.001 (random labels) — a ~1,000× separation.



Evidence, 6 May. Structural velocity per layer, 3-seed mean ± s.d., against scaled training loss.
`v1/original/results/figures/figure1_SV_trajectory.png`

PROCESS NOTES

The language model had to be shrunk to fit the session, so H3 and H5 rested on an 11 M-parameter stand-in. The cross-seed *order* of crystallisation was uncorrelated ($\rho = -0.007$) even though difficulty ordering held — recorded, not explained.

ADJUSTMENTS

Decided to scale every hypothesis to real GPT-2 and to add an **intervention**: detection without intervention is incomplete.

28 JUNE 2026

Version 1 at scale: intervention, six-problem battery, paper

Session 05:52 → 18:48 (~13 h, runs ~5.5 h)
Record: run dirs `20260628_055217`, `_151432`

RESEARCH ACTIVITIES

Morning: a topology-guided bias intervention on pretrained **GPT-2 124M** — detect bias crystallisation, localise the attention heads most aligned with the gender direction, project the direction out *during* training (refreshed every epoch), measure output bias and perplexity; then test whether the precedence window widens with model size (30 M / 51 M / 124 M) and whether the erasure transfers to COMPAS and a toxicity corpus. Afternoon: a 3-hour battery over six problems. Evening: competition paper (17:50) and Quad Chart (18:48).

METHODS & TOOLS

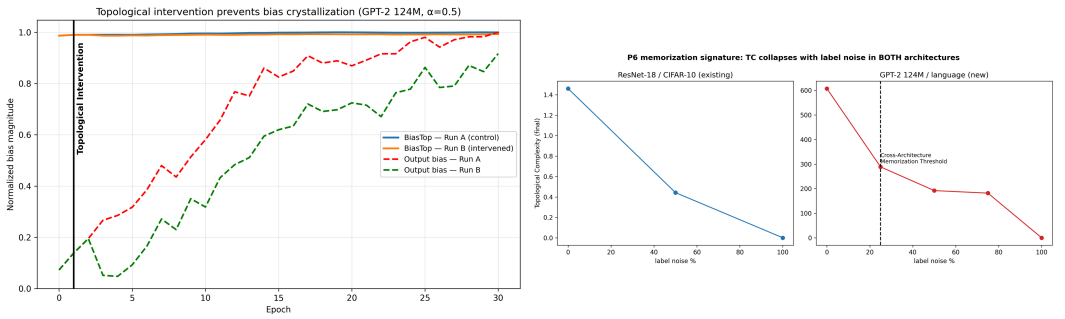
GPT-2 124M with gradient checkpointing; from-scratch GPT-2 configs; ResNet-18; a 4.8 M ViT; Davidson toxicity and COMPAS datasets; Mann–Whitney and permutation tests. Claude Code as execution agent on a clone of the repository, the original left untouched. Targets fixed in advance: ≥ 40% bias reduction at < 2% perplexity cost.

PROCESS NOTES

A one-time, fixed-direction head ablation did nothing; the direction rotates during training, so the erasure has to track it. Applied at block 0 it destabilised the embeddings; blocks 1–3 worked. Only $\alpha = 0.5$ cleared the target ($\alpha = 0.3$: 32% / +0.9%; $\alpha = 0.1$: nothing).

STAGE OUTCOMES

POSITIVE gender-stereotype bias −70.8% at +1.9% perplexity (controlled). **NULL** precedence window at all three scales (permutation p 0.30–0.97). **NEGATIVE** COMPAS: 6.8% gap reduction at −2.3% accuracy; toxicity gap worsened. TC separation ~2,900× on GPT-2. FPL vs per-class accuracy $\rho = -0.84$ ($p = 0.002$). Failure neighbourhoods $p = 8.2\times10^{-6}$. **REFUTED** H5: the ViT shows no mid-training SV peak.



Evidence, 28 June. Left: output bias with and without the intervention. Right: the TC "memorisation collapse" — the figure that carried the v1 paper and that the August nulls would take apart.
`results/whitebox_breakthrough/…/INT-1_headline.png`, `results/whitebox_3hour/…/A_TC_vs_noise.png`

ADJUSTMENTS	Recorded two honest negatives (universality; real-world transfer) in the same document as the positives, and re-scoped P3 to "controlled settings only". **Did not run a shuffle null on TC** — noted two months later as the central omission of version 1.

2 – 29 JULY 2026

Papers, the national round, and the first literature check

Duration ~11 h across sessions (est.)
Record: PDFs 2 Jul, 6 Jul, 29 Jul

RESEARCH ACTIVITIES	**2 July** — regenerated every figure and table from saved arrays. **6 July** — NAS-track paper, "WHITEBOX: A Topological Framework for Developmental Interpretability of Neural Network Learning" (10 pages) with Quad Chart. **~8 July** — searched the literature for prior work on every claimed "first". **29 July** — rebuilt the paper and its source from scratch.
METHODS & TOOLS	LaTeX to the competition format; figures at 300 dpi from run outputs; arXiv and proceedings search.
PROCESS NOTES	Figures in the first draft had been reconstructed from summary numbers rather than raw arrays — caught and fixed before submission. The literature check found the framework-level space already occupied: Neural Persistence Dynamics (NeurIPS 2024), Rieck et al. 2021, HOLE (2025), Rucco 2026.
STAGE OUTCOMES	Paper and Quad Chart submitted; **the project won the US national round.** Several "firsts" downgraded; the intervention and the developmental framing were the least crowded.
ADJUSTMENTS	Two standing rules: every figure must be traceable to a results file; every claim is re-searched for novelty at write-up time and neighbours are cited, not re-claimed. On 12 August the code, results and paper were published on GitHub; on 27 August an explainer deck was written for a general audience.

PART II · 28 – 30 AUGUST 2026

The controls arrive

Three days in which the project's best-known result was shown to be an artefact — and the method that governs everything afterwards was adopted.

28 – 29 AUGUST 2026

Library build with proper nulls

Session 28 Aug 22:58 → 29 Aug 02:12 commits
~12 h agent time · 3 h review (est.)

RESEARCH ACTIVITIES

Rebuilt WHITEBOX as a Python library, `whitebox-topo`: audit and skeleton; fast topology at scale; a calibrated memorisation signal; model-vs-data correspondence; validation batteries.

METHODS & TOOLS

giotto-ph with GPU distance matrices; bootstrap confidence intervals; **feature-shuffle** and **label-permutation** nulls; Representation Topology Divergence for correspondence; GPT-2 124M → XL benchmarks; a 31-model calibration battery (21 CNN, 10 LM) trained with graded label noise. Claude Code as execution agent.

PROCESS NOTES

Feature-shuffled GPT-2 activations kept 77–86% of raw TC: the statistic was measuring how spread out the activations were, not their loops. Landmark sparsification preserved TC ordering but destroyed β_1 (shipped, unused).

STAGE OUTCOMES

> *The 2,900× collapse was an artefact. Scrambled-label models never fitted the noise; their activations simply contracted, and $TC(\lambda X) = \lambda^2 TC(X)$. Models that genuinely memorise show no collapse.*

What survived: a per-family signal (leave-one-seed-out AUC 0.98 CNN / 1.00 LM, converged checkpoints only) whose sign *flips* between CNNs and LMs. All three correspondence hypotheses refuted. P4 dies under a label-permutation null and loses to softmax confidence per sample (0.54 vs 0.90). P1 does not replicate on full-data models. P5 null.

ADJUSTMENTS

Treated the collapse as the project's founding event rather than a setback. From here on every claim is **pre-registered and paired with the null that would kill it**. The v1 headlines were retired with their mechanisms written down.

~29 AUGUST 2026

Literature campaign: what actually makes a black box

Duration ~4 h (est.)
No file record; date recalled

RESEARCH ACTIVITIES

Systematic review of what frontier labs cannot see inside their own models.

METHODS & TOOLS

Méloux et al. 2025 (non-identifiability of explanations); Adolfi et al. 2025 (NP-hardness and Σ_2^p-completeness of circuit queries); Engels, Smith & Tegmark 2024 (SAE "dark matter"); published circuit-tracing limitations.

STAGE OUTCOMES

Seven opacities reduced to three mathematical roots — **gauge** (coordinate multiplicity), **gluing** (local-to-global), **structure** (tractability) — and one gap: no existing method certifies its findings against the network's own symmetry group.

ADJUSTMENTS

The next phase attacks roots, not symptoms.

30 AUGUST 2026

Six pre-registered experiments in a day; “The Structure of Understanding”

Session one day, 11 commits
~5 h authors · agent runs in parallel (est.)

RESEARCH ACTIVITIES

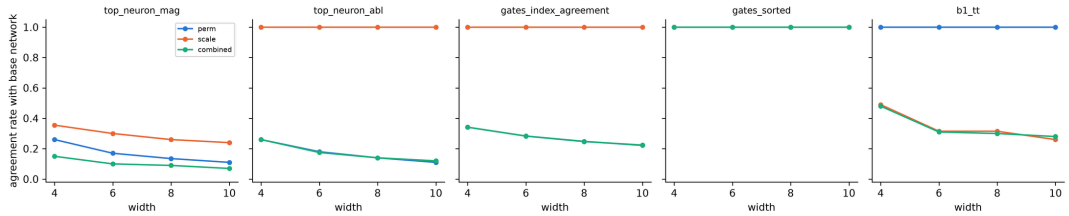
Proposed 26 mathematical experiments (frame-theoretic packing bounds, tensor identifiability, gauge orbits of explanations, invariant observables, a treewidth phase diagram, a Marchenko–Pastur null on SAE residue, sheaf cohomology of explanations). Ran novelty searches on the four flagships, then executed six — E1 to E6 — each with criteria fixed before computation.

METHODS & TOOLS

687 small networks for the gauge experiments; 87 models for the legibility phase diagram; sparse autoencoders for E6; the fly head-direction connectome for E5. Claude Code as execution agent; instructed to finish without pause points but to keep the kill-tests.

STAGE OUTCOMES

FAIL E1 explanation multiplicity is *not* gauge (0/56). PASS E2 gauge-invariant observables exist and the TC scaling exponent is exactly 2. FAIL E3 explanation cohomology. FAIL E4 legibility law, $\rho(Q, S) = -0.37$. FAIL E5. PASS / FAIL E6: the Marchenko–Pastur spectral null passes, the topological half fails. Every *predictive* claim failed its pre-registration; four *instruments* survived and shipped into the library (invariants, legibility, dark matter, gluing).



Evidence, 30 Aug (E2). Agreement of five “explanations” with the base network under permutation, rescaling and combined gauge transformations. Only sorted gate patterns are invariant; top-neuron explanations fall to 10–30%.
`v2-programme/E2/results/fig_agreement.png`

PROCESS NOTES

Sheaf-of-explanations: already taken (arXiv 2603.17141). Modularity-aids-interpretability: qualitative version taken. The gauge-orbit bridge between parameter-symmetry theory and explanation non-identifiability: open.

ADJUSTMENTS

Wrote “The Structure of Understanding” and an operating constitution: *pre-register; kill before believe; negatives are deliverables; re-check novelty; result beats framing.*

PART III · 1 – 17 SEPTEMBER 2026

The gauge programme

Certify explanations against the symmetry group of the parameterisation. A large, disciplined sprint whose most lasting product was a short proposition about why one of its own tests could not fail.

1 – 7 SEPTEMBER 2026

Sprint design and the Agent Team Handbook

Duration ~6 h (est.)
Record: handbook 7 Sept; 11 commits

RESEARCH ACTIVITIES	Scoped the sprint to what is open: a **mechanistic invariance audit** (extending the implementation-invariance axiom of Sundararajan et al. 2017 from input attributions to internals), cross-seed uniqueness, the dark-matter spectral null, the legibility phase diagram. Second, stricter novelty pass on every test.
METHODS & TOOLS	An Agent Team Handbook for execution: shared infrastructure S1–S5, experiment blocks, mandatory pause points, a novelty board. Long jobs run as systemd units behind a GPU queue.
STAGE OUTCOMES	A 65-test roster; execution begun on a **3,648-network Boolean battery**. Per-concept memorisation detection and attribution invariance downgraded from claims to baselines.
ADJUSTMENTS	Made explicit that topology is the concept underneath — invariants of a group action, components of a quotient, gluing obstructions — with persistent homology as one instrument among several.

8 – 9 SEPTEMBER 2026

The within-network tautology: K/N = 1

Duration ~3 h (est.)
Record: 51 commits; pause docs 0–3

RESEARCH ACTIVITIES	Reviewed the Tier-0 report: symmetry backends verified to 10^{-14}; 3,648 networks trained (86% converged); all minimal explanations enumerated; the orbit partition returned K/N = 1.0 *everywhere*.
PROCESS NOTES	A result that is identical in every cell is a statement about the test. For generic θ the stabiliser $\mathrm{Stab}_G(\theta)$ is trivial, so two sub-circuits of the *same* network are never gauge-related: the test could not have returned anything else. The planted "N = 2K" control had been swapped out because recombined duplicate circuits are all valid — which is itself the finding.
STAGE OUTCOMES	> *Proposition. Explanation multiplicity has two sources: coordinate multiplicity across parameterisations (gauge) and structural redundancy within one (not gauge). Uniqueness must therefore be tested across seeds.*
ADJUSTMENTS	Primary metric amended to cross-seed circuit alignment; K/N = 1 kept as a documented finding.

10 – 17 SEPTEMBER 2026

Scale-out, mathematical write-ups, and two paper drafts

Duration ~14 h authors (est.)
Record: 58 commits; documents 14 Sept; claims board 17 Sept

RESEARCH ACTIVITIES	Ran the remaining blocks on the laptop while staging the heaviest for university HPC and cloud GPUs. Certified the GPT-2-small IOI and greater-than circuits on 20-sample gauge orbits (14

	Sept, 1.2 h on the laptop GPU). Wrote a plain-language overview and a mathematical-foundations document; drafted two papers with pending cells marked.
METHODS & TOOLS	Realisation map and its fibres; gauge group $G = \prod_\ell S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}$; functional dimension; path products as exact invariants; Čech construction for gluing; Marchenko–Pastur; parameterised complexity. SLURM templates, a Colab notebook, systemd GPU queues.
PROCESS NOTES	Much of this week went to logistics rather than mathematics — a firewall blocking the cluster from home, a cloud subscription with zero GPU quota, enumeration jobs that were CPU-bound for days. A 49-model drift study finished; the largest enumeration cells did not. Four open questions were posed for a topologist: completeness of G, uniqueness up to gauge, generators of $\mathbb{R}[\Theta]^G$, fixed-parameter tractability in a width parameter.
STAGE OUTCOMES	Two foundations documents; two paper drafts; a claims board on which only instruments and refusals — no predictive law — had survived.
ADJUSTMENTS	Recognised that the programme had grown a defensive apparatus — gauge, sheaves, random-matrix theory, complexity — that buried what the project was for. It had begun defending claims and hunting for novelty instead of using mathematics to open the box.

PART IV · 19 – 21 SEPTEMBER 2026

Blinding Light: one object, four runs

Everything prior was archived intact. The project restarted from an empty directory with a single geometric object and a rule: a result must be a *recovery against ground truth*, not a correlation.

19 SEPTEMBER 2026

Reset: a Riemannian theory of representations

Duration ~6 h (est.)
Record: archive note 19 Sept; specification PDF

RESEARCH ACTIVITIES

Restated the project on one object: the activation manifold M_ℓ at each layer carrying the metric **pulled back from the output**, $g = J^\top J$ with $J = \partial(\text{logits})/\partial h_\ell$; the maps between layers; the bundle over training time. Gave each of the seven problems a definition on that object: generalisation as geodesic closure of class fibres (P6); the fibre inventory with circular coordinates from H^1 (P1); the geodesic explanation (P2); a spurious feature as a cohomology class with trivial task pull-back, plus an **interrupt loop** that penalises it during training (P3); tubular-neighbourhood ignorance (P4); curvature × Lipschitz failure (P5); birth order across checkpoints (P7).

METHODS & TOOLS

A 10-hour specification with numeric pass criteria, nulls and baselines for every cell, written before any code. Systems with **known ground truth**: a transformer learning addition mod 113 (its learned Fourier frequencies are knowable), a ring-attractor RNN (a circle), rotated MNIST (a circle per digit), ResNet-18, GPT-2.

PROCESS NOTES

Literature check: circular-coordinate recovery on network internals was published in June 2026; early shortcut detection is taken (Prediction Depth, SPARE); *intervening* on a detected class during training appeared open. Instruments are treated as dependencies; the problems are the aim.

ADJUSTMENTS

Scoring rule for everything that follows: verdicts **by the letter** of the pre-registered criterion. Anything decided after seeing data is labelled POST-HOC and can never upgrade a verdict. The word “complete” is retired.

19 SEPT 23:33 → 20 SEPT 05:45

Run 1 — the ten-hour specification

Run 6 h 12 min wall-clock · ~2 h review (est.)
Record: `RUNLOG.md`, hourly notes

RESEARCH ACTIVITIES

Built the library (`whiteboxtopo`: manifold, topology, geometry, problems, `open()`), trained every system, computed all seven cells on each, and scored them.

METHODS & TOOLS

Jacobians by autograd; geodesics by Dijkstra on the g-weighted kNN graph; Ripser; de Silva–Morozov–Vejdemo-Johansson circular coordinates; isotypic decomposition by DFT over the group parameter; integrated gradients, softmax confidence/margin and Euclidean kNN as baselines. Claude Code as execution agent, running unattended overnight; every job a systemd unit.

```
def geodesic_distance_matrix(fit):
    """All-pairs geodesic distance on the pullback-weighted graph ..."""
    D = dijkstra(fit.W, directed=False)      # edge weights  w_ij = sqrt( Δh^T (J^T J) Δh )
    ...
    return np.maximum(D, D.T)                # → Rips on D : cohomology of (M, g), not of (M,
Euclid)
```

PROCESS NOTES

The first session died at 00:56 when its terminal tab closed; resumed 01:07 and every long job was moved to a detached unit. Launching too many jobs at once (load average 76–189, 30 GB RAM) got one killed for memory — fixed with capped BLAS threads and sequential queues. A parallel test-writing pass (143 tests) found three real library bugs, including a noise band with a 50% false-positive rate; all fixed, pre-fix outputs kept on disk.

STAGE OUTCOMES

PASS **P2**: path-product attribution equals integrated gradients at $r = 1.0000$ on 100% of inputs; margin CV 0.07–0.10. PASS speed: `open()` 4.6 s per layer. PASS / FAIL P1 on Z113: 5/5 learned frequencies at $r = 0.91$–0.996 with the group action supplied; unsupervised Rips fails (70–207 unnamed classes, best $r \approx 0.2$). PASS P3 detection: shortcut flagged within 0–4% of training on 5 seeds, 0 false alarms in 205 clean and decoy checkpoints. FAIL P4 everywhere — on GPT-2 *below chance* (0.40 vs softmax 0.92). FAIL P5; FAIL P6 by the letter.

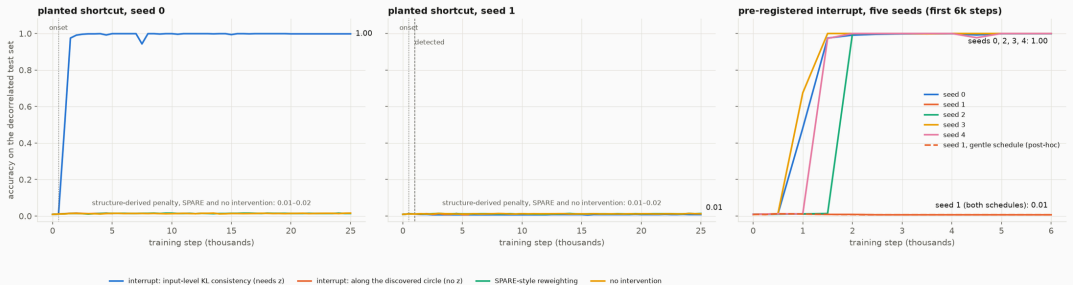
The pullback metric is the right metric for cohomology. A ring RNN's hidden state is only ~50% heading; in Euclidean distance it is a thick blob (coordinate $r = 0.75$–0.80). Under the g-geodesic distance the same cloud is one clean circle, 17–20× longer-lived than any other class, tracking heading at $r = 0.98$–0.99 — exactly the network's own read-out accuracy. *The metric quotients out what the output does not read* (post-hoc).

The specification's ground truth was wrong, and the network said why. Answer-fibres $\{(a, c-a)\}$ were expected to be circles. They are intervals: the model is commutative, so within a fibre each frequency carries $e^{ika} + e^{ik(c-a)} = 2e^{ikc/2} \cos k(a - c/2)$, a standing wave. The fibre is $S^1/\mathbb{Z}_2$. Axis ratio 0.016–0.025 in the fibre, 0.996–0.999 on an unfolded slice.



Evidence, Run 1. Grokking on Z113: geodesic closure rises with the birth of the frequency circles — as a ramp, not a step, and it keeps tracking loss spikes after accuracy reads 1.00. `figures/grokking.png`

> *The evasion result. A penalty derived from the detected circle destroyed the circle — and the shortcut re-encoded itself in a non-circular form. Three consecutive checkpoints: zero detected classes, shortcut reliance 1.000. A monitor of a representation's shape, used as a training signal, can be satisfied by a re-encoding that leaves the function unchanged.*



Evidence, Run 1. Interrupt loop across five seeds: a KL-consistency penalty that knows the nuisance variable recovers 4/5; the structure-derived penalty is evaded. `figures/interrupt.png`

ADJUSTMENTS

"Geodesic closure cannot be faked" is false at a late layer: a metric derived from the output makes "near" mean "same output". Wrote the Run 2 brief to attack each diagnosed cause.

20 SEPTEMBER 2026 · 09:24 → 13:15

Run 2 — the Morse-theoretic upgrade

Run 3 h 51 min wall-clock (budget 7 h)
Record: `SPEC2.md`, `RUNLOG2.md`, 4 pause docs, amendments A-1…A-6

RESEARCH ACTIVITIES

Seven blocks: depth profiles of fibre connectivity and intrinsic dimension; kernel displacement for ignorance; a Morse–Smale complex of the max-logit with separatrix distance for failure; functional alignment to catch the evasion; sensitivity-ranked cohomology; a vineyard across training; a plateau score.

METHODS & TOOLS

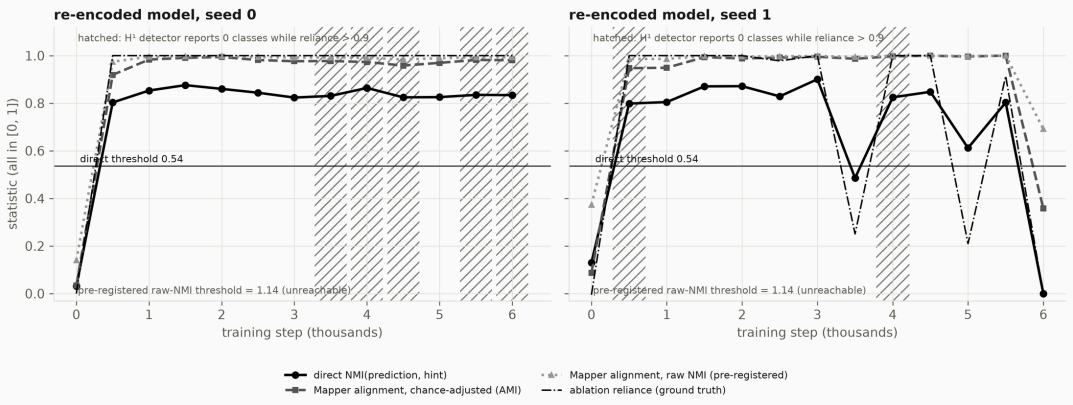
Fibred barcodes for two-parameter persistence; Mapper graphs compared by NMI/AMI; discrete Morse–Smale on the manifold graph; Wasserstein-matched vineyards; TwoNN intrinsic dimension; Gromov–Wasserstein distortion. Unseen models named in advance.

PROCESS NOTES

The registered Mapper threshold came out at 1.139 — unreachable for a statistic bounded by 1; anticipated in a timestamped amendment *before* the data. Chance-corrected, it false-alarms on 44/102 decoy checkpoints: it detects that a nuisance is *represented*, not that it is *used*. Eleven processes on one 8 GB GPU killed two seeds; rerun.

STAGE OUTCOMES

FAIL By the letter no block passed in full. MET The $\mathbb{Z}_2$ quotient replicated on two unseen models. The evasion reproduced step for step from a seed. What *does* catch the re-encoded model is plain dependence of the prediction on the hint over a decorrelated probe — a **functional floor**: 7/7 evasion checkpoints, 0/153 false alarms. POST-HOC floor trigger + Run 1's penalty: 5/5 seeds at $\lambda = 3$ in one round; every escalation past that destroyed the model.



Evidence, Run 2. The H^1 detector goes silent on the re-encoded model while the functional floor keeps firing. `figures/run2_alignment_vs_h1.png`

ADJUSTMENTS

Shape can be re-encoded; function cannot. Last-layer geometry ties softmax because the last layer is linear. Seen after the fact and set aside for proper pre-registration: fibre connection scale falls with depth in generalisers only; the intrinsic-dimension profile has three shapes. **Reframe:** what a network knows is a *symmetry group* — the null directions and isometries of g on input space. LieGG (Moskalev, Sepliarskaia, Sosnovik & Smeulders, NeurIPS 2022) identified as the nearest neighbour and cited in every findings file from here on.

20 SEPTEMBER 2026 · 13:38 → 15:40

Run 3 — the symmetry group

Run 2 h 02 min wall-clock (budget 3 h)
Record: `SPEC3.md`, `RUNLOG3.md`, A-7…A-12

RESEARCH ACTIVITIES

Seven instruments, each verified on planted truth before touching a network: finite automorphism search of the g-distance matrix; null space of g; Killing fields by Lie-derivative residuals; a symmetry-breaking score (share of $\|J\|^2$ on a nuisance slot); quotient depth; orbit explanations; fresh-seed replications trained *after* the criteria were written.

METHODS & TOOLS

Isometry search by 3-anchor Hungarian assignment with all-anchor refinement and verification; characters by DFT along the discovered generator's own coordinate, matched up to one unit of $\mathbb{Z}_{113}^{\times}$. 202 tests, 1 documented xfail.

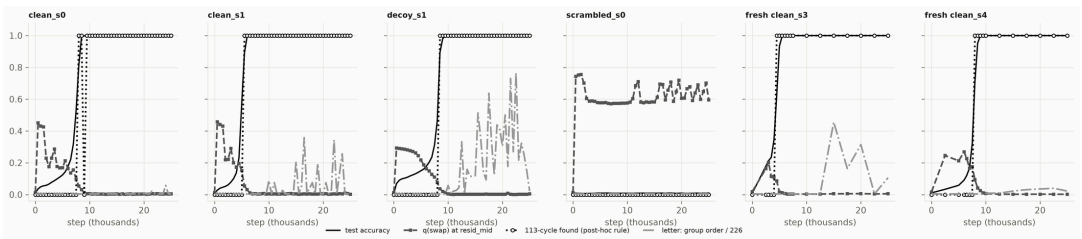
PROCESS NOTES

Planted truth caught two instrument faults before any network was analysed: a tolerance looser than the point spacing accepted 2,712 "isometries"; the 3-anchor assignment lost 221 of 226 true ones. Both fixed by amendment. The brief said: "if the group is not discoverable without being supplied, say so first." By the letter it was not — entrywise isometry finds orders 1, 1, 2, 13, 5 — because a trained network's symmetry is *statistical*: a true translation moves D by 3% on average but 14–42% in its worst entry. A first post-hoc mean-norm rule passed 804 permutations on an *untrained* model and was thrown out the same half-hour.

```
A-10 · 2026-09-20 14:17 · POST-HOC  The mean defect is measured against chance: δ_rand = mean
defect of 10 random
permutations; accept σ iff mean|D_σ - D| < 0.25·δ_rand.  The only meaningful read-out is whether
a FULL n-cycle
is among the accepted permutations.  (Fixed before the fresh seeds finished training.)
```

STAGE OUTCOMES

POST-HOC A full 113-cycle is found on 5/5 generalisers and on no memoriser, within one checkpoint of grokking; its characters recover 5, 4, 4, 4, 4 of the five learned frequencies **with no group supplied**. Swap-invariance is present at initialisation, *broken* during memorisation (0.20–0.46) and *restored* at grokking. On decoy models the nuisance slot holds 44–48% of the trace of $\bar{g}$ during memorisation and 0.04% after. PASS fresh seeds: connection scale falls with depth in 4/4 generalisers (slopes −0.57 to −0.49); ID-profile shapes 20/20 bootstraps. FAIL Killing dimension is 0 for clean, memorising and untrained alike. NOT SCORED interrupt (GPU contention).



Evidence, Run 3. Group formation across training: the generator appears at grokking.
`figures/run3_group_formation.png`

ADJUSTMENTS

The ResNet "memorisers" had only 56–59% training accuracy — they cannot support any memorisation claim, and every one made on them was withdrawn. **Synthesis — understanding as quotient:** collapse of $\bar{g}$ onto the task, preservation of architectural symmetry, a recoverable group; memorisation as the refusal to quotient. Run 4 brief: true memorisers first, then test the synthesis properly.

20 SEPT 15:59 → 20 SEPT 23:49

Run 4 — the tower of quotients

Run 7 h 50 min wall-clock, incl. 4 h 20 min lost idle
Record: `SPEC4.md`, `RUNLOG4.md`, A-13, 3 pause docs

RESEARCH ACTIVITIES

A sanity gate (memorisers must reach ≥ 99% train accuracy at chance test accuracy), then: five memorisation instruments; group discovery with the Run 3 post-hoc rule now **pre-registered** and tested on four fresh seeds; the tower $q_\ell(T)$ of quotients through depth; three formation signals; a dominant-direction rule with a null that could fail; the interrupt, finally scored.

METHODS & TOOLS

True memorisers: ResNet-18 on a fixed 5,000-image subset with random labels, no augmentation, no weight decay; Z113 with random labels and decoy hints. Slot trace-shares of $\bar{g}$; participation-ratio concentration $C = 1 - \mathrm{PR}/\mathrm{PR}_{\mathrm{untrained}}$; shift consistency; dominant-direction flag at slot share > 0.8. The brief's own citation errors were corrected in the specification before scoring. 206 tests.

```
$ python analysis/demo_run4.py
open(Z113 fresh generaliser (clean, seed 7))
  group: generator FOUND (statistical symmetry, full 113-cycle) = translation by 1; characters
at [14, 12, 21, 34, 45]
  tower q(T) embed→resid_mid→resid_post→logits:  swap 0.77 0.00 0.01 0.01;  hint_change 0.00
0.02 0.04 0.04
  instruments: slot shares (a, b, hint) = (0.50, 0.49, 0.01);  q(swap) = 0.004  → 0/3 on the
memoriser side   [12s]
open(Z113 true memoriser (random labels + decoy hints, seed 5))
  group: no generator (no full 113-cycle among accepted permutations)
  tower q(T):  swap 0.76 0.55 0.83 0.83;  translation 0.99 0.97 0.98 0.98;  hint_change 0.01
0.38 0.67 0.67
  instruments: slot shares (a, b, hint) = (0.35, 0.38, 0.28);  q(swap) = 0.547  → 3/3 on the
memoriser side   [46s]
```

Evidence: terminal transcript, `logs/demo_run4_transcript.txt`. Nothing about $\mathbb{Z}_{113}$ is supplied to the instrument.

PROCESS NOTES

The laptop GPU was power-capped by the system to roughly a sixth of its clock, so training ran ~6× slower than planned; the ResNet job was paused so six Z113 runs could finish (A-13). **A mistake worth recording:** the final-model analysis crashed at 17:15 on a one-line autograd error, and the supervising agent was waiting for a log line that a crashed job could never print. Four hours idle, 17:15–21:35, found only when the authors asked how much longer. New rule: wait on process exit, never on a log line. Twice, numbers were written into a findings file before being checked; both were corrected against the results files and the corrections kept.

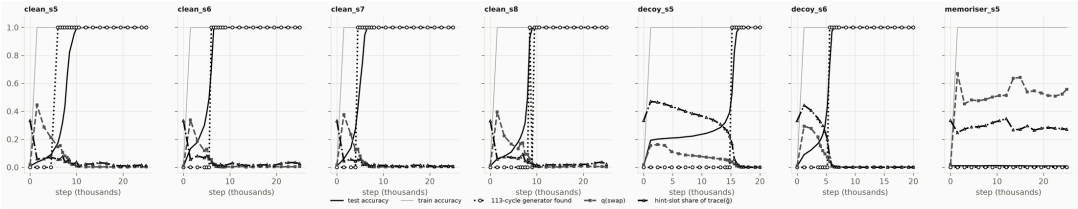
STAGE OUTCOMES

Gate passed: ResNet memorisers 99.6 / 10.4 and 99.4 / 10.2 (train / test %); Z113 memorisers 100 / 1.0 and 100 / 0.8.

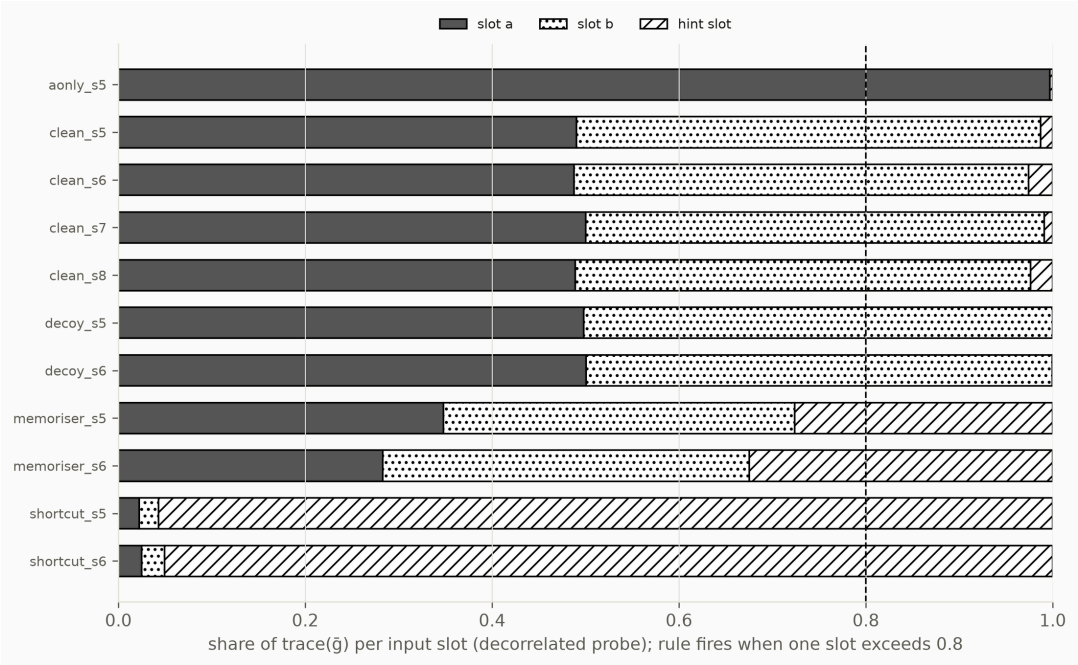
PASS **P1.1 — pre-registered group discovery.** A full 113-cycle on 4/4 fresh generalisers, 0/2 memorisers; 4, 4, 5, 5 of five frequencies recovered at $r > 0.9$, each through a single unit of $\mathbb{Z}_{113}^{\times}$ (1, 33, 1, 20). PASS P1.2 planted truth with a measured operating range. PASS P1.4 orbit explanations 1.000, 1.000, 1.000, 0.990 vs 0.000 for memorisers. PASS **P3.1 dominant direction**: shortcut models put 0.957 / 0.951 of tr $\bar{g}$ on the hint slot; clean models never exceed 0.50; decoys 0.0002–0.0003. PASS P6.2 slot shares: generalisers (0.49, 0.50, 0.01), memorisers (0.35, 0.38, 0.28). PASS P6.5.

FAIL P6.1 concentration (differences +0.02, −0.25, +0.05, −0.03 — no sign). FAIL P6.3 on ResNet: an *untrained* network is the most shift-consistent of all (0.99). FAIL P6.4 (wrong sign). P6 block 2/5. FAIL P1.3: the linearised tower produces artefacts. FAIL P7.1–7.3: the generator appears 0–4 checkpoints *before* grokking, not at it; slot collapse is sharp (0.47 → 0.000) but lands 2–3 checkpoints after.

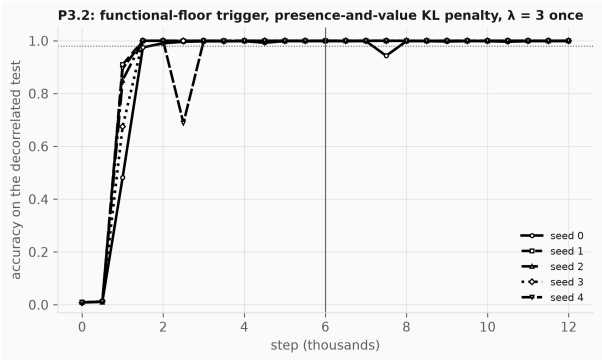
PASS **P3.2 — the interrupt, scored at last (landed 23:49).** Functional-floor trigger (fired at step 500 on every seed), Run 1's presence-and-value KL penalty at $\lambda = 3$ switched on once and never escalated, five seeds, 12,000 steps. Decorrelated accuracy at step 6,000: 0.999 / 1.000 / 1.000 / 1.000 / 1.000. Floor firings in (6,000, 12,000]: 0 in 60 checks. In-distribution test accuracy at 12,000: 0.999 / 1.000 / 1.000 / 1.000 / 1.000. All three registered clauses hold on 5/5 — the cell that was failed in Run 1 (4/5), unscoreable in Run 2 and not scored in Run 3. Recorded although the criterion does not cover it: seed 0 dipped to 0.943 at step 7,500 and recovered by 8,000. The interrupt is not topological and it needs the nuisance variable to be known.



Evidence, Run 4. Formation signals against the grokking checkpoint on fresh seeds. `figures/run4_formation.png`



Evidence, Run 4. The dominant-direction rule: share of tr $\bar{g}$ per input slot for clean, planted-shortcut, decoy and a-only models, with the 0.8 flag line. `figures/run4_dominance.png`



Evidence, Run 4. Decorrelated accuracy for five seeds under the interrupt; vertical line = the registered step-6,000 check. `figures/run4_interrupt.png`

ADJUSTMENTS

What holds after four runs: a generalising network carries a **discoverable group** and a memoriser does not; the **trace of the pulled-back metric** says which inputs a network is using and flags a shortcut without being told what it is. What does not hold: any single scalar "memorisation score", Killing fields, the linearised tower, and formation *timing*. Planted-shortcut models also carry the generator — the group says what was learned, not whether it was learned for the right reason. Next: a non-linearised tower; groups beyond cyclic; the dominant-direction rule on a model where the shortcut is not known in advance.

21 SEPTEMBER 2026 · 13:11 → 22:45

Run 5 — registration

Run 9 h 34 min wall-clock, incl. 5 h 05 min asleep (lid closed)
Record: `SPEC5.md`, `RUNLOG5.md`, A-14, 3 pause docs

RESEARCH ACTIVITIES

A confirmation run: nothing new proposed. The numbers Run 4 had seen on four matched pairs were written down as thresholds, eleven new models were trained, and everything was scored by the letter — four instruments plus the group for the definition of understanding, four cells for the generator as a leading indicator, and a finite (Jacobian-free) tower to replace the linearised one.

METHODS & TOOLS

SPEC5 saved at 13:12:51, training launched at 13:14, with eight things Run 4's files already said about the brief written down first (a near miss on file, nulls known in advance not to separate, sparse-sweep leads). ResNet clean models trained *with* random-crop-and-flip augmentation, memorisers without. Every checkpoint swept (51 per model). Finite g-length of the segment between activations, verified on planted truth (3 tests). Label-permutation null for the depth slope. 209 tests pass.

PROCESS NOTES

The laptop lid was closed at 14:00 and the machine slept until 19:05 with eight analysis processes mid-flight; they resumed intact and nothing was recomputed, but five hours were lost. The GPU ran at full clock (2,350 MHz), so all eleven specimens trained in 36 minutes.

STAGE OUTCOMES

PASS q(swap) 0.004–0.006 vs 0.47–0.68; shift consistency 0.98 vs 0.18–0.20; group found 3/3, never on a memoriser or an untrained network. FAIL slot collapse on one memoriser (hint slot 0.445 vs a 0.42 ceiling); depth slope on one ResNet memoriser (−0.47 vs −0.30) — whose label-permutation null is −0.30: on a CNN the slope mostly reads whole-cloud contraction, on the transformer it reads what it claims (null −0.11 vs −0.53). P6 block 3/5; all 12 clean-side clauses hold, 17/19 overall. PASS the generator is accepted at the largest accuracy jump on 2/3 seeds and one checkpoint before on 1/3, at test accuracy 0.51–0.68, never after and never on 102 memoriser checkpoints; restoration and decoy hint collapse ($\rho = -0.99, -1.00$) as registered.

FAIL persistence 0.946 vs 0.95 on one seed. PASS finite tower 3/3: a memoriser's hint change is visible at the embedding (0.43–0.48) where the linearisation read 0.00.

ADJUSTMENTS

P6 and P7 stay “found”, not “registered”: two ceilings set from two memorisers failed on the third and fourth, and the gaps are reported rather than moved. Run 4's “the generator leads by 1–4 checkpoints” was a sparse-sweep artefact, said in advance and confirmed. The finite tower is registered. Next: memoriser-side clauses of the right shape (no collapse, not a ceiling on the maximum); the depth slope with its null subtracted on CNNs.

21 SEPTEMBER 22:52 → 22 SEPTEMBER 02:20

Run 6 — orbit-closure

Run 3 h 28 min wall-clock, GPU capped at 30 W
Record: `SPEC6.md`, `RUNLOG6.md`, A-15

RESEARCH ACTIVITIES

Asked the group to do two more jobs. **Knowledge as orbit coherence**: $\kappa(x)$ = the fraction of the model's own group on which $f(g{\cdot}x) = \rho(g)f(x)$ — a network knows an input iff its learned rule holds there. **Failure as equivariance defect**: $\varepsilon(x) = \max_g \|f(g{\cdot}x) - \rho(g)f(x)\|$. Scored against max-softmax, entropy, ODIN and Golan & El-Yaniv's 72 designer transformations on thirteen models trained after the thresholds were written; closed the definition of understanding with the memoriser clause re-registered in the right shape.

METHODS & TOOLS

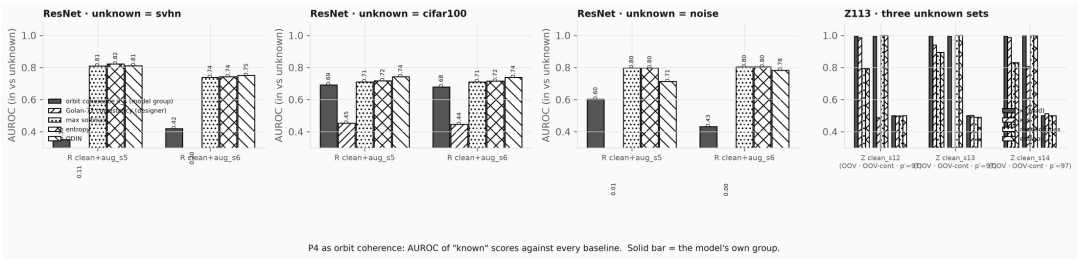
`whiteboxtopo/knowledge.py` (κ, κ_s, ε, $\bar{\varepsilon}$; the Z113 group with answer relabelling, $|G| = 7$; the CNN's shifts and flip, $|G| = 13$; Golan-72; a non-group null), four planted tests. Unknown sets: OOV tokens and random embedding vectors for Z113; SVHN, CIFAR-100 and Gaussian noise for the CNN. Failure radius by PGD-direction bisection. SPEC6 wrote nine things down in advance, including that the "different modulus" set is a subset of valid inputs and that an untrained CNN already has its architecture's symmetries. A-15 fixed the brief's failure clause before analysis: the softmax margin's correlation with the failure radius is *positive*, so "more negative than" is vacuous.

PROCESS NOTES

The GPU was power-capped at 30 W (80 W requested) for the whole run; training took 2.1 h instead of 36 min. Not changed. 213 tests pass.

STAGE OUTCOMES

PASS The definition closes on the transformer: re-registered slot collapse 6/6 with the untrained null on the memoriser side, q(swap), the group, and null-subtracted fibre tightening (−0.39 to −0.45 vs −0.05). FAIL On the CNN, null-subtracted, memorisers' fibres tighten exactly as much as generalisers' (−0.12 vs −0.11 to −0.13): the instrument is transformer-only. PASS Orbit coherence finds an out-of-vocabulary Z113 input at AUROC 0.994–0.996 against softmax's 0.79–0.89, and memorisers know nothing by it ($\kappa = 0.00$ / 0.23). FAIL On the CNN it is *below chance* on SVHN (0.35, 0.42) and loses to every softmax score: an augmented network is shift-invariant on SVHN, noise and at initialisation alike (κ_s 0.98–0.996). Golan-72 is worse still (0.11–0.45). FAIL The equivariance defect does not predict failure: $|\rho|$ 0.02–0.13 (Z113) and 0.41–0.43 (CNN) against the margin's 0.36–0.53 and 0.93; random non-group transformations do as well. Persistence 0.87 on one seed, whose generator arrived four checkpoints before its jump.



Evidence, Run 6. AUROC of "known" scores against every baseline on the CNN's three unknown sets and Z113's three. `figures/run6_roc.png`

ADJUSTMENTS

The theory is stated for five problems, not seven: a network's knowledge is a group, and that answers what it learned, why it decided, whether it understood, how it formed, and where planted structure lives. It answers what the network does not know only where the group was learned rather than built in, and it does not answer where the network will fail — six runs, three instruments, nothing beats the softmax margin, because the failure radius is defined by descending it. Written down as the closing statement.

22 SEPTEMBER 2026 · 08:07 → 09:30

Run 7 — the extractor, and the end

Run 1 h 23 min of a 5 h budget
Record: `SPEC7.md`, `RUNLOG7.md`

RESEARCH ACTIVITIES

One idea earned by six runs: a learned symmetry is one the trained network keeps and the untrained network does not. Built the extractor that subtracts the architecture — $\lambda_\ell(T) = (q_\ell^{\text{untrained}} - q_\ell^{\text{trained}}) / q_\ell^{\text{untrained}}$ over seventeen transformation families and a ten-dimensional Lie-algebra search — and asked whether its symmetries do on a CNN what the discovered generator did on the transformer. Replaced adversarial failure by natural failure. Rebuilt the transformer from its group alone. Wrote the final claims map.

METHODS & TOOLS

`whiteboxtopo/extractor.py`: rotation, scale, shear, translation, brightness, contrast, gamma, hue, saturation, blur, noise, JPEG, cutout, elastic, shift, flip; three untrained ResNet seeds as the null; contrastive search minimising $q_{\text{trained}} - q_{\text{untrained}}$ over exp(tA). Five CIFAR-10-C corruptions generated with the reference parameters (the dataset was not on disk — said in SPEC7). Three Z113 seeds trained to 12,000 steps and read at the checkpoint nearest 80% test accuracy. Every specimen else reused from Runs 5–6. 213 tests.

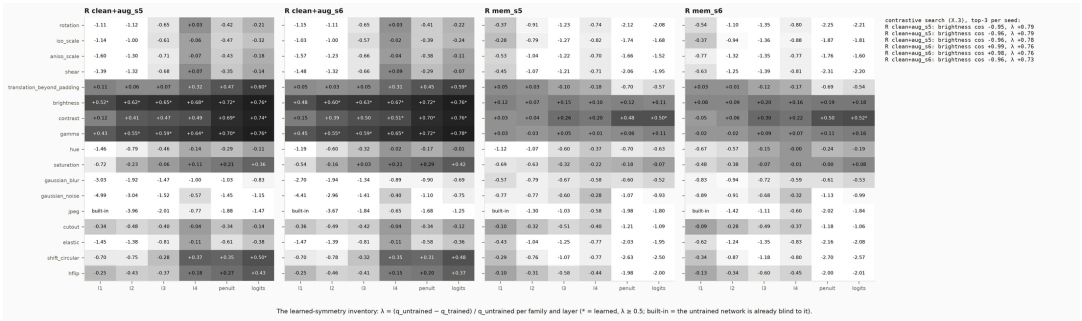
PROCESS NOTES

GPU still capped at 30 W; the whole run's computing fitted in 60 minutes anyway. Seed 18 never reached the 70–90% window and was scored at its nearest checkpoint, as registered.

STAGE OUTCOMES

PASS An augmented ResNet-18 has exactly three learned symmetries — brightness, contrast, gamma (λ 0.69–0.72), none augmented — on both seeds; memorisers have none and are 2–3× more fragile than an untrained net to rotation, scale and shear; the Lie-algebra search converges to brightness unaided (|cos| 0.95–0.99). Training makes the network more sensitive to noise, blur and JPEG (λ −0.6 to −1.9). **FAIL** "Built-in = $q_{\text{untrained}} < 0.05$" is right for the transformer's swap (0.01–0.03; memorisers break it, λ −50 to −76) and wrong for the CNN's shift (0.17–0.30). **FAIL** Learned symmetries lift the CNN's knowledge boundary on SVHN (+0.28, +0.31) and noise, not on CIFAR-100; the cosine-based untrained null fails again (0.997). **PASS** On grokking-in-progress transformers, held-out errors are exactly the orbit-incoherent inputs: AUROC 1.000, 1.000 vs softmax 0.85, 0.90; random transformations 0.44–0.49. **FAIL** On the CNN three photometric invariances do not predict a semantic error (0.67–0.69 vs 0.84). **PASS** The generator and five characters rebuild the transformer at 100% with eleven fitted weights (γ =

0.00); **FAIL** ablation symmetry 0/6 — the network's circuit is a different, less redundant implementation. **PASS** persistence 0.87 / 1.00 / 0.97 ≥ 0.85.



Evidence, Run 7. The learned-symmetry inventory: λ per family and layer on two generalisers and two memorisers, with the contrastive search's top-3. `figures/run7_inventory.png`

ADJUSTMENTS

Stopped. The claims map in `CLAIMS.md` gives every claim its run, evidence file, null, baseline and status. What WhiteBOX is: a network's knowledge is a group — discoverable from outputs on the transformer, extracted by subtracting the untrained network on the CNN — and that group answers what it learned, why it decided, whether it understood, how it formed, and (on the transformer) where it fails and what it has never seen. It does not predict adversarial failure better than the softmax margin anywhere, and on a CNN it marks only the unknown inputs that violate what was learned. Deferred: gluing on GPT-2, extraction on language models, non-abelian discovery, a κ built from the extractor's own units.

20 – 21 SEPTEMBER 2026

Final paper and backboard

Duration ~7 h, in parallel with Runs 2–4 (est.)
Author-reported; no run log

RESEARCH ACTIVITIES

Wrote the final paper, “WHITEBOX: Understanding as Quotient” (21 pages), and the backboard (twelve panels, twenty figures and diagrams, the full test inventory, reserved slots for Run 4).

METHODS & TOOLS

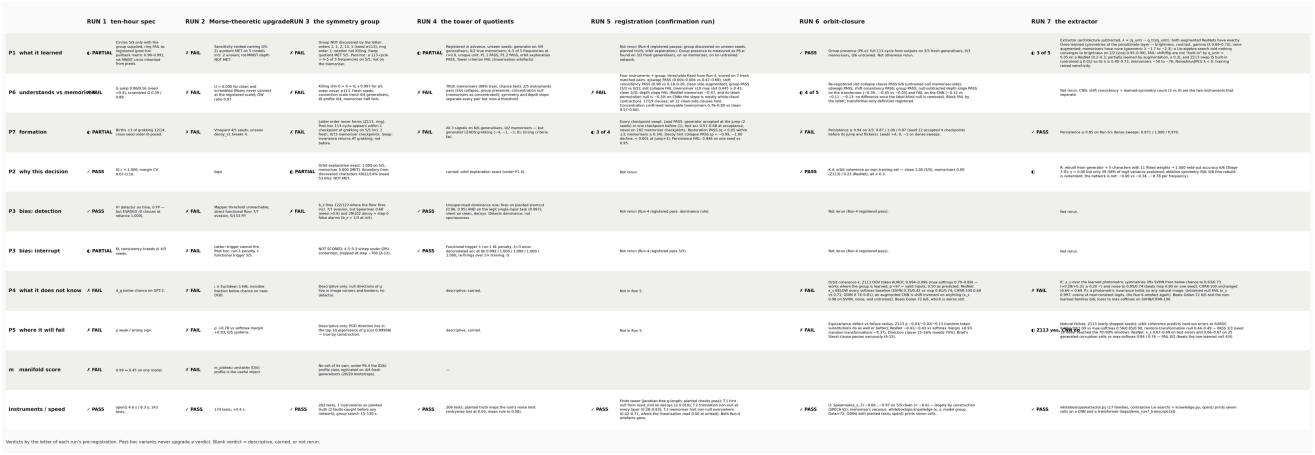
LaTeX, TikZ; every figure converted from run outputs; four published run reports as the source of numbers.

STAGE OUTCOMES

Paper and board produced; the scorecard carries all four runs. Pending cells are filled from run outputs only.

The scorecard, as it stands

Every cell of every run, scored by the letter of a criterion written before the computation. Most cells are red. That is the point of the grid: the green ones mean something.



Seven-run grid. Rows are the seven problems; columns are systems within each run. Generated from the results files by `analysis/make_grid_run7.py`. `figures/grid_seven_runs.png`

Standing methods across all entries since 30 August

RULE	IN PRACTICE
Pre-registration	Numeric pass thresholds in a SPEC file before computation; 13 timestamped amendments across Runs 2–4; unseen models and fresh seeds named in advance.
Planted truth first	Every instrument is run on a synthetic object with a known answer before it touches a network. This caught two instrument faults in Run 3 and three library bugs in Run 1.
A null for every positive	Feature shuffle, label permutation, untrained network, decoy models (“present but unused”), random permutations.
Baselines	Softmax confidence and margin; integrated gradients; Euclidean *k*NN; Prediction Depth; SPARE; LieGG.
Post-hoc is labelled	Anything decided after seeing data is marked POST-HOC and never upgrades a verdict; it becomes a pre-registered cell in the next run or it is dropped. (Run 3’s group rule → Run 4’s P1.1, which passed.)
Traceability	Every figure is generated from a results file by a script in the repository; every number in a findings file names the file it lives in.
Negatives are results	Failures are reported as failures, with the mechanism where one was found. The word “complete” is not used.

Equipment and software. One laptop: NVIDIA RTX 5070 Laptop GPU (8 GB), 30 GB RAM, Arch Linux. Python 3.12–3.14, PyTorch 2.11 (CUDA 12.8), Ripser, persim, giotto-ph, scikit-learn, SciPy, transformers, matplotlib. Datasets: CIFAR-10, MNIST, WikiText-103, WinoBias-style probes, COMPAS, Davidson toxicity, synthetic modular arithmetic, Boolean-function batteries. Pretrained models: GPT-2 (124M–XL), Pythia. AI tooling: Claude Code (Anthropic) as the coding and execution agent under written specifications, from 28 June onward.